

CSE 49 $x|_{x \in \{3, 4\}}$ - Final Year Project

Philosophy, processes and practices

August 4, 2026

Department of Computer Science, School of Mathematics and Computer Science

Contents

0.1	Context	3
0.1.1	Eligibility	3
0.1.2	Learning outcomes	3
0.1.3	The stakeholders	3
0.2	Project tracks	4
0.3	AI-Assisted Development Policy	4
0.4	Project workflow and assessment milestones	5
0.5	Appendices	7
0.5.1	Documentation	7
0.5.2	Title page example	8
0.5.3	INternal/Faculty Advisors	2
0.6	Disedrata	2

0.1 Context

The computing Final Year Project (FYP) is the capstone project of the Bachelor of Science (Computer Science) programme. It puts classroom learning into practice, cutting across the breadth of the degree and serving as a springboard to further mission-based knowledge acquisition and application.

It is a computing core as regulated by the Higher Education Commission of Pakistan, and forms part of the graduation requirement¹ (Table ??).

It is spread across two (2) semesters of study, and comprises six (6) credit hours.

0.1.1 Eligibility

Students are eligible for enrolling/registering in an FYP if they are in their 7th semester of studies in the CS undergraduate programme, and have completed all computing cores offered up to the 6th semester plus 90 credit hours in total.

0.1.2 Learning outcomes

The FYP is meant for students to:

1. identify and elaborate a problem, and formulate and implement a working solution,
2. respond to all segments of the software development life cycle, or alternatively, that of scientific research,
3. use project management methodologies to plan, execute and deploy the solution,
4. exhibit competency in a technology stack, including the ability to justify design decisions, debug and modify source code independently, and reason about system behavior without tool assistance.

0.1.3 The stakeholders

Student group (up to 3), Advisor, Co-advisor(s), the FYP Committee, the OBE Committee, and Alumni mentors, alongside the Program Office, Senior Executive Industrial Linkages, Manager SMCS, and the CS secretariat, as in the previous document.

¹Higher Education Commission of Pakistan, *Curriculum of Computer Science, Software Engineering, and IT* hec.gov.pk.

0.2 Project tracks

The FYP branches into two assessment tracks, as in the original document:

1. **Software track** - a technology-based solution to the proposed requirement.
2. **Research/experimental track** - emphasis on experiment design, method engineering, and reporting of findings, approved in writing with advisor support.

Both tracks share the same five-milestone structure (Section 0.4) and the same milestone names; only what is expected within each milestone differs by track.

	Software track	Research track
M1	MVP	Feasibility Pilot
M2	Internal Evaluation	Internal Evaluation
M3	External Open House	External Symposium
M4	Real-Conditions Validation	Robustness Validation
M5	Final Defense & Report	Final Defense & Report

0.3 AI-Assisted Development Policy

- **Default permission:** code completion tools, AI chat assistance for debugging/explaining errors, boilerplate generation, test-case generation, and documentation drafting are permitted by default, (to be debated) unless an advisor restricts them for a specific project where the algorithm/architecture itself is the intended contribution.
- **Disclosure:** every deliverable (M1, M2/M3, final report) includes a short “AI Tools Used” paragraph - which tools, for what tasks, and roughly how much of the codebase/text they formed.
- **Boundary:** AI-assisted code is generally fine when disclosed; however students ought not to pass AI-generated analysis or reasoning off as their own.
- **Verification:** competency (Learning Outcome 4) is checked directly through the oral/viva components built into the milestones.
- **Resourcing:** where apt, the School may support pay-as-you-go GPU rental costs for compute-heavy projects, on a case-by-case basis.

0.4 Project workflow and assessment milestones

Five core milestones, shared by both tracks. Term 1 covers the MVP and the two closing evaluations of the core build - internal (which also carries the individual-defense component), then external at an Open House; Term 2 covers evaluation/robustness and the final defense.

Term 1 (CSE 493) – Sept-Dec

Milestone	Weeks	Evaluated by
M1 - MVP	~6	Jury + Advisors
M2 - Internal Evaluation	~15	Advisors + Internal Jury
M3 - External Open House	~16 (beginning of new term)	External Jury/Industry

Below describes what will go in every one of the milestones:

M1 - MVP / Feasibility Pilot

Software track: a (fully) working technical demonstrator demonstrating that the hardest or the core part of the idea is feasible (e.g. nontrivial algorithm runs, integration with an external system works), and a written proposal. This will be validated by jury defense; and signed off by the advisor before proceeding.

Research track: a literature review that establishes the research gap and problem context, plus small-scale experimentation (e.g. a trial run of the proposed method on a sample of data) demonstrating the methodology is viable before committing a full term to it. Jury defense; and signed off by the advisor as well.

M2 - Internal Evaluation

Software track: full system functioning end-to-end *against the scope agreed at M1*, reviewed internally by advisors and an internal jury. Includes a live oral walk-through - the jury/advisor picks a component and the student explains or modifies it unaided. Fortnightly advisor review meetings run throughout as a continuous, lightly-graded process.

Research track (suggested): research methodology / experiment pipeline fully implemented and operational, with initial or preliminary results obtained on real or representative data, reviewed internally the same way. Live oral walkthrough: the student justifies a specific methodological choice, or interprets a preliminary result, unaided. The same individual defense applies here too.

M3 - External Open House / External Symposium

The same core build from M2 is presented publicly, this time to an external jury/industry audience rather than internally - called the Open House or the External Symposium. This is also the handoff point: from here on, CEDIR (the umbrella arm

of the Office of Research, Innovation and Commercialization) takes up software-track projects into its own incubation process, running alongside Term 2 rather than as a single end-of-year pitch. For the research track, if CEDIR engagement applies, it likewise begins here rather than at the final defense.

Term 2 (CSE 494) - the extension of the previous process

Milestone	Weeks	Evaluated by
M4 - Real-Conditions Validation / Robustness Validation	~14	Advisors + CEDIR
M5 - Final Internal Defense & Report	~15	Jury + Advisors + Committee + CEDIR

M4 - Real-Conditions Validation / Robustness Validation

Software track: real-world validation - the group proves deployment and working under real conditions (real data/users/load², as scoped with the advisor). Includes security, edge-case, HCI/accessibility and non-functional-requirement review.

Research track: full-scale experiments completed; results validated for robustness - replication, comparison against baselines, and an explicit discussion of threats to validity. Paper written and submitted, with AI reviews available and preprint hosting accepted.

M5 - Final Defense & Report

Software track: final defense with deployed/working software; final report (must document evaluation methodology and design rationale, including AI-tool disclosure). The CEDIR outcome from the M3 handoff is reported here: whether the project was incubated (followed CEDIR's process) or deployed independently, e.g. to the App Store, as its own startup.

Bonus: the system being adopted and actively used by a real organization or by real/paying users beyond the FYP's own testing, not merely hosted somewhere.

Research track: final defense with hypothesis in/validation; final report documenting methodology, results, and a discussion of limitations, with reporting from domain experts or use case validation host for applied projects. (The External Symposium already happened at M3, so this defense can stay focused on the findings themselves.)

Bonus: an additional grade bonus for teams whose FYP research is submitted or published or passes a preprint check, counted against an arXiv preprint or a national/regional CS venue (e.g. INMIC, FIT). Where apt, the School may also

²Is the system live e.g. a non-localhost URL, real infrastructure/environment, and used end-to-end by at least one real user outside the group?

support the costs of a top-ranked conference publication specifically, on a case-by-case basis.

Committee assessment and submission of compliant artifacts to the School are administrative sign-offs attached to this milestone but are equally important.

Documentation checkpoints

Three per group across the year, for both tracks: the M1 proposal report, the M4 evaluation/validation note, and the M5 final report.

The final report and artefacts produced under the FYP are logged for trace and reference at the IBA's library. Their sign off is a graduation requirement.

0.5 Appendices

0.5.1 Documentation

Skeletons for documents may be found linked here.

In addition, here are current forms for evaluating the various milestones milestone 1,
milestone 2,
milestone,
milestone 4,
milestone 5.

These are subject to review.

0.5.2 Title page example

XX YY

Memorable but dorky algorithm....

DOCUMENT NAME E.G. PROPOSAL

Authors:

X
Y
XY
YX

Advisors:

Dr. Awesome advisor

August 20, 2026

Department of



School of Mathematics and Computer Science

The full template for the report and the poster hosted at this repository.

0.5.3 Internal/Faculty Advisors

There is a strong preference for selecting an advisor from the permanent faculty at the CS Department. A local advisor is additionally mandated when working with either an industrial partner or a mentor.

You may find the list of permanent professors below, along with their areas [HERE](#). This list is subject to change.

0.6 Disedrata

The process of the FYP is undergoing active change, and this booklet is subject to evolution and improvement to better reflect to that, as the process itself will change to listen to the voice of all stakeholders including graduates and prospective employers.